

A Walk Through Netfilter

In this article, let's walk through the features of Netfilter, look at how it operates and try out a hands-on programming section to get started with it.



Would it not be useful if we could tap the packets traversing up the Linux kernel stack and handle it our way? Or if we could bypass the standard kernel packet processing and, instead, manipulate the packets to meet our requirements? Fortunately, Linux offers us this capability and flexibility via the sophisticated Netfilter, a versatile framework integrated in the Linux kernel to intercept packets up their way through the kernel stack and process them the way we need to. In fact, Netfilter is the engine behind the famous firewall solution in Linux - the iptables. Netfilter was developed by Paul 'Rusty' Russell and is currently maintained by the Netfilter core team. Here is what we can do with Netfilter:

- Inspect and filter the packets.
- Implement stateless and statefull firewalls.
- Perform network address and port translation (NAPT)
- Alter the packets (TOS/DSCP/ECN fields in IP headers)
- Implement transparent proxies and a lot more.

The anatomy of Netfilter

Any packet accepted by the NIC will travel up the network stack to the application if the packet is addressed to the host machine or will otherwise get forwarded to the next hop (if a Linux box acts as a router). Making the decision to send the packet to the application on the local host or route it to the next hop happens at Layer 3, where Netfilter operates. Netfilter defines a bunch of 'openings' in L3 of the TCP/IP stack in the kernel called 'hooks'. Kernel modules can register at any or all of these hooks to listen to and gain access to packets. The kernel modules, in turn, invoke a call-back function to custom process the packets and, hence, override kernel packet processing. For the IPv4 protocol, Netfilter defines five hooks as depicted in Figure 1.

1. **NF_INET_PREROUTING**—All incoming packets, after the sanity check, pass this hook.
2. **NF_INET_LOCAL_IN**—All packets destined to the local host go through this hook.